The State Tone Index: A Multidimensional Framework for Measuring Macro-Social Affective Climate and Institutional-Emotional Conditions

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Abstract

This paper introduces the State Tone Index (STI), a conceptual-methodological framework for organizing indicators of macro-social affective climate and institutional-emotional conditions at the country level. The paper responds to a measurement gap: existing comparative indicators separately measure wellbeing, governance, rights, trust, social cohesion, press freedom and fragility, but rarely provide a transparent construct map linking these domains.

The proposed framework draws on literature on emotional climate, collective emotions, social cohesion, social trust, institutional trust, legitimacy and composite indicators. STI is treated as a multidimensional framework rather than a direct measure of a single national emotion. The current v0 pilot uses accessible public proxy indicators from the World Happiness Report, Worldwide Governance Indicators, Freedom House and Reporters Without Borders, with Transparency International CPI retained as a contextual source outside the baseline score.

The paper contributes a construct map, indicator-role logic and reproducible pilot workflow. The exploratory v0 demonstration separates wellbeing/agency, governance/openness and rights/information dimensions, then tests whether baseline rankings are sensitive to alternative weighting and exclusion scenarios. The demonstration shows why decomposition and rank-sensitivity review are more defensible than a single headline country ranking.

The current STI v0 is not validated and does not directly measure emotional climate. It excludes direct social-trust survey variables, emotional-climate scale items, media-sentiment measures, conflict-event data and direct polarization indicators. The pilot is therefore best interpreted as a proxy-condition and

methodology demonstration. Future work must add direct construct indicators, test convergent and discriminant validity, expand uncertainty analysis and evaluate temporal robustness.

Keywords: State Tone Index; emotional climate; collective emotions; social cohesion; social trust; institutional trust; composite indicators; governance; press freedom; comparative politics.

1. Introduction

International comparisons of national condition usually separate economic performance, subjective wellbeing, governance quality, civil liberties, press freedom, social trust and social conflict into different measurement systems. This separation is analytically useful, but it makes it difficult to describe how institutional conditions, social cohesion and collective affect interact as a broader state-level climate (OECD & Joint Research Centre, 2008; Schiefer & van der Noll, 2017).

Research on emotional climate and collective emotions shows that societies can be studied through shared affective orientations and group-level emotional processes (Bar-Tal et al., 2007; De Rivera & Paez, 2007; Goldenberg et al., 2020; von Scheve & Ismer, 2013). Related literatures on social cohesion, social trust, institutional trust and legitimacy show that the emotional and relational condition of a political community is not reducible to income, regime type or one institutional score (Chan et al., 2006; Levi & Stoker, 2000; Schiefer & van der Noll, 2017). At the same time, composite-indicator methodology warns that aggregation without explicit construct boundaries, normalization, weighting and sensitivity analysis can create misleading rankings (OECD & Joint Research Centre, 2008; Saisana et al., 2005).

Existing indicators capture important fragments of national condition, but they rarely provide a transparent framework for connecting macro-social affective climate with cohesion, trust, legitimacy, information openness and governance conditions. The gap addressed here is therefore not the absence of data, but the absence of a reproducible construct map and decomposition workflow that can organize these fragments without presenting them as a single validated diagnosis.

This paper introduces the State Tone Index (STI) as a conceptual-methodological framework for organizing measurable indicators of macro-social affective climate and institutional-emotional conditions. The current v0 pilot measures proxy conditions around state tone rather than emotional climate directly. The contribution is fourfold:

- define state tone as a multidimensional construct rather than a literal national emotion;
- map emotional climate, collective emotions, social cohesion, trust, legitimacy, rights, information openness and governance into distinct construct layers;
- propose a transparent composite-index workflow with explicit normalization, weighting and sensitivity requirements;
- demonstrate an exploratory v0 pilot as a decomposition exercise, not as a validated ranking.

The research question is: Can macro-social affective climate be operationalized as a transparent multidimensional framework that integrates emotional climate, collective emotions, social cohesion,

social trust, institutional legitimacy and information-governance conditions?

2. Literature Review

The first foundation for STI is the literature on emotional climate and collective emotions. Paez et al. (1997) and De Rivera and Paez (2007) support the idea that emotional climate can be treated as a social and cross-cultural object of study rather than only as an individual psychological state. Bar-Tal et al. (2007) connect collective emotions to societal processes in conflict situations, while von Scheve and Ismer (2013) and Goldenberg et al. (2020) provide broader theoretical and review-level support for collective emotions as group-level phenomena.

For STI, this literature justifies the conceptual layer of macro-social affective climate. It does not, by itself, validate a country-level composite score. This paper therefore uses the literature to justify the construct and separately treats measurement as an explicit methodological problem.

The second foundation is social cohesion. Chan et al. (2006), Schiefer and van der Noll (2017), and Fonseca et al. (2019) show that social cohesion has been defined and operationalized in multiple ways, usually involving connectedness, belonging, shared orientation, cooperation and social relations. STI treats social cohesion as a distinct construct layer. Schiefer and van der Noll's (2017) three-part organization around social relations, identification with a geographical unit and orientation toward the common good is especially useful for this framework. Fonseca et al. (2019) add a multi-level view that connects individual belonging, community relationships and institutional inclusion. STI does not equate social cohesion with political conformity, absence of dissent, equality, quality of life or simple regime stability.

The third foundation is trust. Putnam (1995) frames social capital and civic connectedness as important properties of societal functioning. Hu (2017) and Justwan et al. (2018) add measurement caution by showing that trust is not a single simple quantity: generalized trust, radius of trust and aggregate trust measures can differ conceptually and empirically. For STI, future WVS or equivalent survey variables need careful construct mapping. Generalized trust, radius of trust, confidence in institutions, social support and interpersonal trust should not be merged without construct justification.

The fourth foundation is institutional trust and legitimacy. Levi and Stoker (2000) review political trust and trustworthiness. De Cremer and Tyler (2007) connect trust in authority and procedural fairness to cooperation, and Hamm et al. (2017) develop an integrated framework linking fair process, trust and legitimacy. Valgardsson et al. (2025) adds a current cross-national frame for institutional-trust trends. STI separates institutional trust and legitimacy from social trust and from objective governance capacity.

The methodological foundation is OECD and Joint Research Centre (2008) and Saisana et al. (2005). Composite-indicator construction requires transparent construct definition, indicator selection, normalization, weighting, imputation, robustness and sensitivity analysis. STI follows this logic by emphasizing construct boundaries, dimension scores, source limitations and rank sensitivity rather than a single headline score.

Table 1. Evidence Summary

Claim / need	Evidence	Status	Use in manuscript	Caveat
Emotional climate can be treated as a social construct	Paez et al. (1997); De Rivera and Paez (2007)	Supported	Conceptual foundation	Does not validate STI v0
Collective emotions can emerge at group/social levels	Bar-Tal et al. (2007); von Scheve and Ismer (2013); Goldenberg et al. (2020)	Supported	Macro-affective premise	Requires scale/level distinction
Social cohesion has measurable dimensions	Chan et al. (2006); Schiefer and van der Noll (2017); Fonseca et al. (2019)	Supported	Cohesion layer	Definitions differ
Social trust requires careful operationalization	Putnam (1995); Hu (2017); Justwan et al. (2018)	Supported	Future WVS/trust layer	Not included in v0
Institutional trust and legitimacy are distinct from social trust	Levi and Stoker (2000); De Cremer and Tyler (2007); Hamm et al. (2017); Valgardsson et al. (2025)	Supported	Institutional-emotional layer	Current v0 uses proxies
Composite indicators require transparent construction and sensitivity analysis	OECD and Joint Research Centre (2008); Saisana et al. (2005)	Strongly supported	Methodology	Must be implemented, not only cited

Note: Table 1 summarizes manuscript-relevant evidence. It does not imply that STI v0 is validated; it identifies the literature and methodology supporting the framework.

3. Methodology

This study is a conceptual-methodological working paper with an exploratory data demonstration. It proposes a State Tone Index framework and demonstrates a reproducible pilot workflow, but it does not claim validation, prediction or policy diagnosis.

STI is defined as a multidimensional framework for macro-social affective climate and institutional-emotional conditions. The construct map separates emotional climate, collective emotions, social cohesion, social trust, institutional trust, legitimacy, information openness, rights conditions, governance capacity and wellbeing/agency. In this version, social cohesion is treated as a relational construct involving social relations, belonging/identification and orientation toward the common good; social trust is separated from institutional confidence; and legitimacy is treated as connected to institutional trust and procedural fairness.

The current v0 pilot uses only accessible public proxy variables. It contains three practical dimensions:

1. Wellbeing/agency: life evaluation, social support and freedom to make life choices from the World Happiness Report (World Happiness Report, 2025).

2. Governance/openness: government effectiveness, rule of law and voice/accountability from WGI (World Bank, n.d.).
3. Rights/information: political rights, civil liberties and press freedom from Freedom House and RSF (Freedom House, n.d.; Reporters Without Borders, 2025).

The v0 pilot excludes direct emotional-climate scale items, WVS trust variables, V-Dem variables, ACLED events, GDELT/media sentiment and direct polarization indicators. These exclusions are made explicit because they define the interpretive boundary of the current demonstration.

Table 2. Indicator Roles in STI v0

Dimension / role	Indicators in v0	Source	Role	Caveat
Wellbeing / agency	life evaluation; social support contribution; freedom contribution	World Happiness Report 2025	baseline proxy dimension	Adjacent to state tone, not direct emotional climate
Governance / openness	government effectiveness; rule of law; voice/accountability	World Bank WGI	baseline proxy dimension	Expert/perception-based governance estimates
Rights / information	political rights reversed; civil liberties reversed; press freedom score	Freedom House; RSF	baseline proxy dimension	Composite/normative indicators
Corruption context	CPI score	Transparency International	contextual only	Not in baseline score
Stability/conflict context	WGI political stability; future ACLED	contextual/future	excluded from v0 baseline	Conflict-event extraction pending
Trust/cohesion	future WVS variables	future core candidate	excluded from v0	Wave alignment unresolved
Democracy/polarization	future V-Dem variables	future candidate/control	excluded from v0	Manual access and variable selection unresolved
Media/emotion signal	future GDELT/media sentiment	experimental candidate	excluded from v0	Bias/noise methodology unresolved

Note: STI v0 uses proxy dimensions only. Direct trust, cohesion, emotional-climate and media-sentiment measures are not yet included.

Table 3. Pilot Data Coverage and Exclusions

Source group	v0 status	Coverage / issue	Manuscript treatment
World Happiness Report 2025	included	30/30 pilot countries	baseline wellbeing/agency proxy
World Bank WGI/WDI	included	governance/population/R&D mostly complete; researchers per million partial	governance proxy and controls
Freedom House FIW 2025	included	30/30 pilot countries	rights proxy
RSF 2025	included	30/30 pilot countries	information/press freedom proxy

Source group	v0 status	Coverage / issue	Manuscript treatment
CPI 2025	downloaded/contextual	30/30 pilot countries	contextual, not baseline
V-Dem	excluded	manual form/email/privacy consent required	future work / source blocker
ACLED	excluded	API/account/extraction-date decision required	future conflict-event layer
WVS	excluded	wave-level temporal alignment required	future trust/cohesion layer
GDELT	excluded	media-bias/noise methodology required	future media-signal layer

Note: Coverage refers to the current 30-country exploratory pilot. Exclusions are methodological decisions, not claims that excluded sources are irrelevant.

The workflow defines construct layers, maps candidate indicators, excludes circular or unavailable indicators, normalizes variables so that higher values indicate more favorable proxy conditions, computes dimension scores before aggregate scores, compares alternative weighting scenarios and flags rank instability. Exploratory country outputs are interpreted only as method/decomposition examples.

4. Framework Demonstration

The main output is a framework demonstration rather than a validated empirical result. STI v0 organizes country-level proxy indicators into three exploratory dimensions: wellbeing/agency, governance/openness and rights/information. The baseline score averages the three dimension scores. Alternative scores test whether results change when variables are weighted equally, when rights indicators are reduced, or when wellbeing variables are reduced.

The v0 output identifies a high-scoring group in the baseline specification, including Finland, Denmark, Norway, Sweden and New Zealand, and a low-scoring group including Kenya, China, Nigeria, the Russian Federation and Turkiye. These positions are not country diagnoses. They reflect how the selected proxy variables behave under the baseline construction.

Two dimension-divergent cases, Mexico and the United Arab Emirates, were flagged by rank-sensitivity checks. Mexico appears more favorable on wellbeing/agency than on governance and rights/information proxies. The United Arab Emirates appears stronger on governance/wellbeing proxies than on rights/information proxies. These cases support the argument that decomposition is necessary: a single score can obscure dimension-specific patterns.

Table 4. Exploratory STI v0 Sensitivity Summary

Group	Countries	Interpretation
Top 5 baseline proxy scores	Finland; Denmark; Norway; Sweden; New Zealand	Exploratory baseline behavior under selected proxy variables
Bottom 5 baseline proxy scores	Kenya; China; Nigeria; Russian Federation; Turkiye	Exploratory baseline behavior under selected proxy variables

Country	Baseline rank	Max rank shift	Interpretation boundary
Mexico	20	5	Dimension-divergent; wellbeing/agency stronger than governance and rights/information
United Arab Emirates	23	6	Dimension-divergent; governance/wellbeing stronger than rights/information

Note: All values are exploratory and non-validated. Table 4 demonstrates decomposition and sensitivity logic, not country performance or national emotional condition.

5. Discussion

The STI v0 demonstration supports a narrow methodological interpretation: country-level proxy indicators can be organized into a multidimensional framework that separates wellbeing/agency, governance/openness and rights/information conditions. The most important finding is not the rank order itself, but the behavior of the framework under decomposition and alternative weighting choices.

The divergent cases identified in the v0 sensitivity review illustrate why a single composite score is not enough. Countries may appear relatively stronger on one dimension and weaker on another. This is central to the STI argument: state tone should be interpreted through a profile of conditions, not through a headline score.

The framework is consistent with literature that treats emotional climate and collective emotions as social phenomena, but the current pilot does not directly measure them. It also draws from social-cohesion and trust literatures, but current v0 data do not yet include direct WVS or equivalent trust variables. Methodologically, the approach follows composite-indicator guidance by making indicator selection, weighting and sensitivity review explicit.

The main limitations are that STI v0 measures proxy conditions, not emotional climate directly; direct trust, cohesion, legitimacy, polarization, media-sentiment and conflict-event variables are excluded; several input sources are themselves composite or perception-based indices; WGI, Freedom House, RSF and CPI may overlap conceptually; weighting and normalization choices remain subjective; no external validation has been conducted; and cross-cultural and language biases remain unresolved.

6. Conclusion

This paper proposes the State Tone Index as a multidimensional framework for organizing measurable proxy conditions around macro-social affective climate. The contribution is not a validated ranking of countries. The contribution is a construct map and transparent workflow that show how candidate indicators can be assigned to conceptual layers, normalized, decomposed and tested for sensitivity.

The next research step is to strengthen the construct validity of STI by adding direct trust, cohesion, legitimacy and emotional-climate indicators, then testing whether STI dimensions converge with

related measures and remain distinguishable from existing governance, wellbeing and rights indices. Only after this validation work should STI be presented as more than an exploratory framework.

Declarations

Funding: This research received no external funding.

Conflict of interest: The author declares no conflict of interest.

Author contributions: Rim Zabarov is the sole author and is responsible for conceptualization, methodology, analysis, writing and final submission decisions.

Data and code availability: The current pilot uses public aggregate country-level datasets. Source metadata, extraction notes, hashes, interim extracts, the processed pilot panel and analysis scripts are stored in the project workspace. Data and code may be made available by the author upon reasonable request, subject to third-party dataset terms. A clean public data/code package may be released in a later version after redistribution and licensing terms are reviewed.

Ethics approval: The current v0 pilot uses public aggregate country-level data and does not involve direct human-subject data collection. Future use of social-media data, media sentiment, conflict-event microdata or sensitive group-level inference requires ethics and data-use review.

AI assistance disclosure: AI assistance was used to organize research materials, verify sources, draft working notes and prepare manuscript sections. The author remains responsible for source verification, interpretation, analysis, final text and submission decisions.

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